

MODULE 1 – Sequencing & QC

Answers to module questions

- 1. Remembering the fastq file format, how many reads do you have represented in the previous box?**

Two

- 2. How many paired reads are in the sample?**

2115140 CNGB1553_S31_L001_R1_001.fastq → $2115140/4 = 528785$

CNGB1797: 1288460

CNGB39120: 1718971

- 3. Does the number of reads in R1 and R2 files match?**

Yes

- 4. Analysing the FastQC reports, how many reads are there? Does this answer match your previous answer for sample CNGB1553 (based on `wc` command)?**

528785. Yes

- 5. What is the average depth of coverage for these samples??**

CNGB1553: 53X

CNGB1797: 123X

CNGB39120: 102X

- 6. With respect to quality scores, is the quality of read 1 exactly the same as that of read 2? Do you see the same in all the samples?**

No, quality of read2 is usually worse than read1. Yes, it's a pattern that repeats in all the samples

- 7. What is the most abundant species identified in each sample?**

CNGB1553: *Escherichia coli*

CNGB1797: *Staphylococcus aureus*

CNGB39120: *Mycobacterium tuberculosis*

- 8. Are samples contaminated with other bacterial species?**

CNGB39120: 25.51% reads *Staphylococcus epidermidis*

- 9. How many paired reads are left in the sample after trimming? Compare them with the fastq files obtained from the sequencer**

CNGB1553: 506450

CNGB1797: 1264306

CNGB39120: 1666587

10. How many reads are there in CTMA_1441_longds.fastq.gz? What is the median read length, N50 and median quality score?

Reads: 90000

Median read length: 1983

N50: 16061

Median quality score: 9.4

11. What is the most abundant species identified in this sample?

Vibrio cholerae

12. Examine the results for the filtered file and try to answer the questions you answered for the unfiltered file. Do you find any difference among these results?

Reads: 43795

Median read length: 7931

N50: 16974

Median quality score: 9.8